

Computer Animation Technology in Behavioral Sciences: A Sequential, Automatic, and High-Throughput Approach to Quantifying Personality in Zebrafish (*Danio rerio*)

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Computer animation technology in behavioral sciences: A sequential, automatic and high-throughput approach to quantifying personality in zebrafish (*Danio rerio*)

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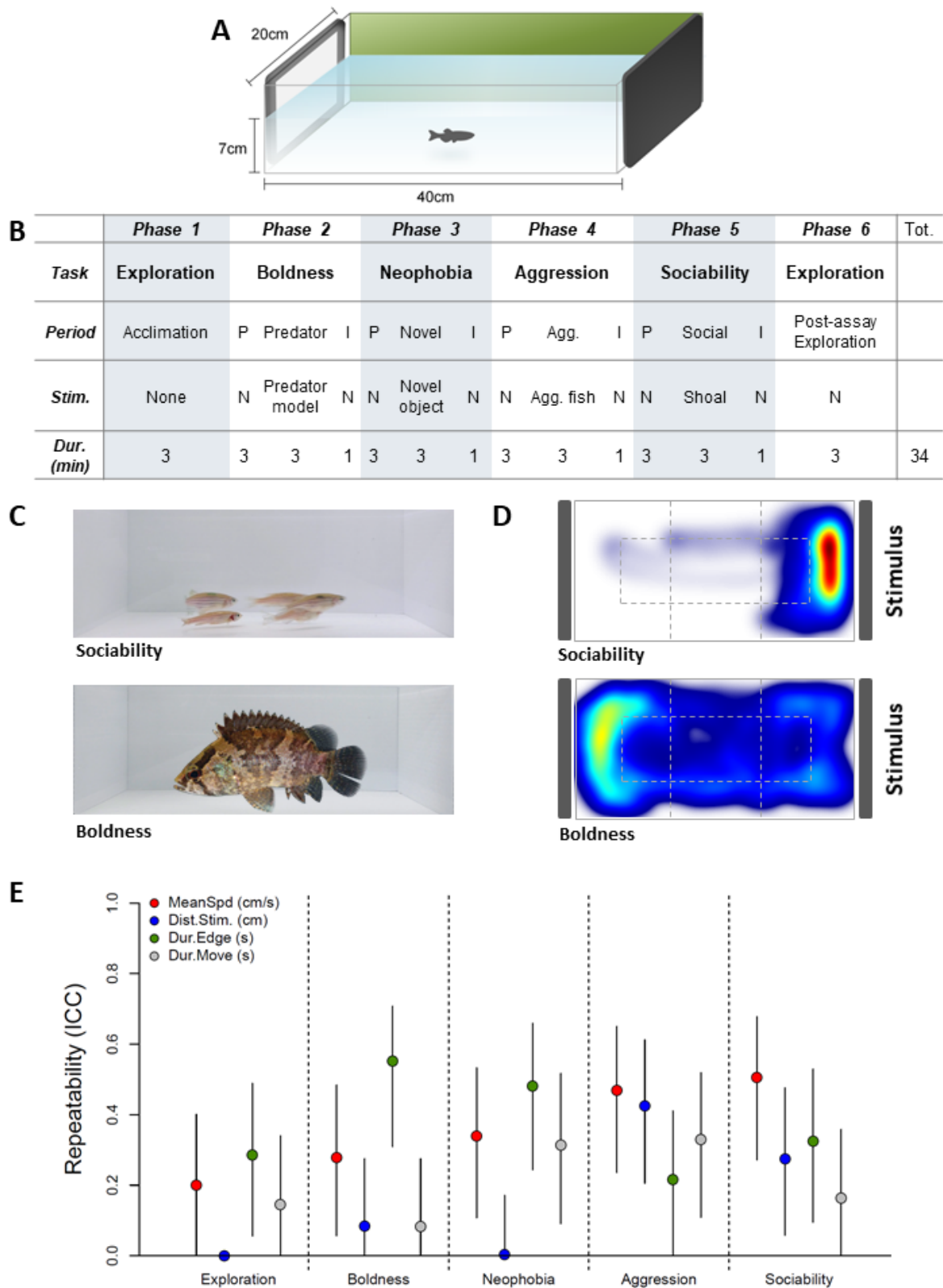
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Running Head: A high-throughput method to quantify personality



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19 **Figure 1: A.** The experimental setup. Each tank had a white base to heighten subject contrast, and dark green
20 walls. The nearest wall has been omitted for the visual representation of the inside of the tank. Two 8-inch
21 computer tablets were placed against the transparent short walls of the tank, one blank (no stimulus, control) and

the other displaying the stimulus video. **B.** A timeline of tasks digitally presented to subjects, across a 34-minute period. Stimulus videos were divided into six phases. Phases 1 and 6 are the exploration phases, where zebrafish are exposed to a neutral (N, no stimulus, control) background image. Phases 2 to 5 consist of four digitally simulated assays: boldness (exposure to an animated predator model), neophobia (exposure to an animated novel object), aggression (exposure to a video of a single aggressive zebrafish), and sociability (exposure to a video of a shoal of zebrafish). Phases 2 to 5 are pseudorandomized across trials, generating 24 unique trial combinations. **C.** Representative frame from stimulus videos. The sociability assay consisted of a video of a zebrafish shoal (5 fish), while the boldness assay consisted of an animated predator model swimming across the stimulus screen. **D.** Heat maps generated by EthoVision for the sociability assay and boldness assay, demonstrating typical behavioral responses (distributions of fish location) that are elicited by these two stimuli. **E.** Intra-class correlations (repeatabilities) and their 95% confidence intervals for four behavioral measurements (mean speed during movement in tank, mean distance of the fish to the stimulus, duration spent in the outer edge zone of tank, and duration moving in the tank) for the five behavioral assays associated with personality (exploration, boldness, neophobia, aggression and sociability).

Abstract

An emergent field of animal personality necessitates a method for repeated high-throughput quantification of behavioral traits across contexts. Here, we have developed an automated video stimulus approach to sequentially present different contexts relevant to five ‘personality’ traits (exploration, boldness, neophobia, aggression, and sociability), successfully quantifying repeatable trait measurements in multiple individuals simultaneously. Although our method is designed to quantify personality traits in zebrafish, our approach can accommodate the quantification of other behaviors, and could be customized for other species. All digital materials and detailed protocols are publicly available online for researchers to freely use and modify.

Keywords: Behavioral phenotyping, automation, animal personality, consistency

Methods to rapidly quantify behavioral traits in model animal species are becoming increasingly important in the fields of neuroscience, behavioral sciences and evolutionary biology¹. Among behavioral traits, characterizing and quantifying ‘personality’ traits are especially time-consuming due to the number of repeated measures that need to be taken to demonstrate individual trait repeatability (i.e., significant intra-class correlation, ICC^{2,3}; for a recent work on personality in zebrafish, see⁴). Fortunately, reduced costs and advances in computer technology permit not only repeated behavioral measurements of single traits, but also the measurement of suites of behaviors, in a completely automated, standardized, customizable and high-throughput manner without observer bias⁵. For example, tablet computers can present videos or animations to animals to elicit a behavioral response, while detailed behavioral measurements can be obtained with automated tracking software from video recordings. These types of methods have been explored previously in zebrafish^{1,6-13}, but the application of multiple behavioral assays associated with personality in one continuous assay has not yet been implemented.

Here, we have developed and trialled, using twenty-four adult zebrafish (male = 12; female = 12, approximately 6 months old), an automated video stimulus approach that sequentially presents behavioural contexts relevant to five personality traits: 1) exploration (movement in a novel environment), 2) boldness (response to a predator), 3) neophobia (response to a novel object), 4) aggression (response to a single aggressive conspecific), and 5) sociability (response to a shoal of conspecifics). Behavioral trials occurred between 1 and 4 pm, over four days, at weekly intervals, measuring each individual four times over the course of the study (i.e., a subject was trialled once a week). For each trial, fish were placed individually into an experimental tank (40 x 12 x 20 cm, length x depth x width) filled to a water depth of 7 cm. Tanks were constructed from 6mm acrylic consisting of a solid white floor, dark green long walls and transparent short walls. On either side of the short walls, an electronic tablet

(Windows 10 operating system, 8-inch screen, 1280 x 600 resolution) was positioned flat against the clear acrylic, projecting a computer-generated image to the experimental zebrafish (Figure 1A). Thermostat-controlled heat mats (Ultimate Heat Mat, 40W, 40 x 60 cm) maintained water temperature at ~28°C for the duration of the trial.

Each trial lasted for 34 minutes, in which a single video consisting of 6 phases (Figure 1B) was presented to the zebrafish. Phases 1 and 6, the ‘exploration’ phases, consisted of a 3-minute neutral background image (no stimulus, control). Phases 2-5, the ‘stimulus’ phases, consisted of a 3-minute pre-stimulus (no stimulus, control) period, a 3-minute stimulus period, and a 1-minute interval (no stimulus). Each stimulus phase comprised one of four tasks: social preference, novel object exposure, aggression, or predator avoidance (Figure 1A, 1C). These tasks were designed to elicit a specific behavioral response (Figure 1C; Supplementary material). Phases 2-5 were pseudorandomized, creating 24 possible video combinations. Trial videos were presented on either the left or right hand side of the tank, with the opposite tablet presenting a neutral (no stimulus) background image for the trial duration. Each zebrafish was randomly assigned to a tank and exposed to four unique videos and tanks, experiencing the stimulus on both the left and right hand sides of the tanks in a balanced manner.

Trials were filmed from above using a Panasonic HC-V770M camcorder (720p resolution). All videos were analysed using EthoVision XT 11.5 automated tracking software. Here, we present repeatabilities intra-class correlation, ICC; ³; for the four behavioral measurements (mean speed, mean distance to the stimulus, duration in the outer tank zone or edge, and duration moving) during the five personality assays (Figure 1E). We found one or more significantly repeatable measurements, whose effect size was moderate to high, in each assay. These repeatable behavioral measurements can be used to compare individual or group differences, or look at across context correlations. Using the methods described herein (with the use of 8

tanks), up to 64 individuals can be trialled within a 6-hour period by one operator, allowing higher-throughput designs to be implemented.

Notably, although we did find repeatable behavioral measurements under different contexts, it is premature to conclude that these five assays could completely capture five different personality traits¹⁴. Detailed analyses of different behavioural measurements under varied contexts are likely required to profile different animal personality traits. As somewhat expected¹⁵, our ‘boldness’ and ‘neophobia’ assays produced similar repeatability patterns in the behavioural measurements shown in Figure 1E. As did the ‘aggression’ and ‘sociability’ assays. However, in both cases, the degree of repeatability in the duration of moving could be a separating factor to distinguish different contexts (Figure 1E). As pointed out before, however, we also recommend more than one assays to characterise a personality trait when possible^{16,17}. Additional, analysis profiles in Ethovision may allow for more detailed behaviours to be characterised that better distinguish the different contexts (e.g., the number of zone transitions from the stimulus for characterising aggressive behaviour).

Importantly, all our stimulus videos, associated files, templates and detailed protocols can be found online (Supplementary material; doi 10.17605/osf.io/rga6e). Therefore, researchers can use or customise our method with the use of available templates, depending on their specific experimental needs. For example, the 1-minute interval we used between stimulus presentations can be easily prolonged to reduce a potential carryover effect although such carryover effects can also be statistically dealt with¹⁸. Although we have designed our method using zebrafish, our approach can be easily re-designed for any other species, expanded to include more assays and used with new, freely available, animal tracking software e.g., ¹⁹.

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Author Contributions

SN, DH and DN conceived ideas and developed them with inputs from all the others. MF, DN and SN designed the methodology. MF designed and generated all stimulus videos. MF performed the experiments with preparatory help from RO, TU, ML, DH and DN. DN analysed the data. MF wrote the manuscript with input from SN and DN. MF wrote and collated all supplementary material with assistance from ML, RO and DN. All authors contributed critically to the drafts.

Disclosure Statement

The authors declare no conflict of interest.

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Computer animation technology in behavioural sciences: A sequential, automatic and high-throughput approach to quantifying personality in zebrafish (*Danio rerio*)

Supplementary Information

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1. Supplementary templates and stock videos

S1: [Boldness assay](#)

A .zip folder containing a collection of five predator models in .png and animated .gif format,
a .ppt motion animation template, and five motion animated videos.

To utilise the motion animation template, open the .ppt file. To substitute predator models,
right click on the predator model and select *Change Picture*. Save as a Windows Media Video
(.wmv).

S2: [Neophobia assay](#)

A .zip folder containing a collection of 11 novel objects in .png format, a .ppt motion
animation template, and 22 motion animated videos.

To utilise the motion animation template, open the .ppt file. To substitute novel objects, right
click on the novel object and select *Change Picture*. Save as a Windows Media Video (.wmv).

S3: [Aggression assay](#)

A .zip folder containing a collection of ten 5-minute video clips of an aggressive stimulus (an
individual adult zebrafish performing aggressive behaviours).

S4: [Sociability assay](#)

A .zip folder containing a collection of thirteen 5-minute video clips of a social stimulus (a
shoal of five adult zebrafish).

S5: [Stimulus video composition](#)

A .zip folder containing a Windows Live Movie Maker template, a Photoshop CS6 template, and accompanying video tutorials demonstrating their use.

S6: [Stimulus videos](#)

A .zip folder containing 24 compressed stimulus videos, as used in this experiment.

S7: [Trial data collection](#)

An .xls template for preliminary data collection.

S8: [EthoVision template](#)

This template includes arena settings, trial control settings, data profiles and analysis profiles utilised in this experiment. Included are also sample videos, a sample trial list, and descriptions of the data and analysis profiles used. Additional more detailed settings that were not analysed in this experiment have also been included (additional arena zones, data profiles and measured parameters) for exploratory purposes. This template should serve as a starting point to navigate EthoVision according to the associated written protocol provided (see S12), and should be adapted according to specific experimental needs.

2. Supplementary protocols

Protocols have been written for use in the present study and may include details specific to our laboratory and equipment. Protocols have been supplied as a guide only, and may be adapted as required.

S9: [How to: Film aggression and sociability stimulus videos](#)

A written protocol detailing the methods used to select and film zebrafish for aggression and sociability contexts.

S10: [How to: Zebrafish personality trials](#)

A written protocol detailing the methods used to perform zebrafish personality trials. These methods outline setting up the experimental tanks, setting up the video camcorders and computer tablets, and trial execution. In these methods, Camera 1 was used to film tank group 1 (tanks 1-4), and Camera 2 was used to film tank group 2 (tanks 5-8).

S11: [Video tutorial: Zebrafish personality trials](#)

A video tutorial demonstrating how zebrafish personality trials were conducted for this experiment. This includes setting up the computer tablets and trial execution. To be viewed in conjunction with the written protocol (see S10).

S12: [How to: EthoVision](#)

A written protocol detailing the use and functionality of EthoVision as used in this experiment. This protocol outlines all steps from creating a new experiment to data acquisition for the present study. Included are instructional figures and troubleshooting tips to maximise successful data acquisition. To be used in conjunction with the supplied EthoVision template (see S8).

3. Detailed behavioural phenotyping protocol

3.1. Stimulus video composition

3.1.2. Required Windows programs

- 1) Microsoft PowerPoint 2016
- 2) Photoshop CS6, or alternative image editing and animating software
- 3) Windows Live Movie Maker (Windows Live Essentials, 2012), or alternative video editing software
- 4) VLC Media Player, or alternative video compressing software

3.1.3. Animating predator models and novel objects

Predator model species were selected based on their use in previous studies^{1, 2}. Five different photographs of leaf fish (genus *Nandus*) were sourced and the body of the fish was isolated onto a transparent background using Photoshop CS6 (Figure S1A). Waveform animation was added to simulate a semi-realistic swimming effect using the *Filter > Distortion > Wave* function over a series of at least twenty frames. Frames were compiled using the *Window > Timeline > Create Frame Animation* function at a frame rate of 0 seconds until an effective swimming motion was achieved (Figure S1B). Files were exported as .gif with matte colour selected to blend into the chosen background image. Isolated .png files and animated .gif files of each of the five predator models are available online (see S1).

Twenty-two novel objects were designed and modified using the *Shapes* tool in Microsoft PowerPoint and saved as transparent .png objects. Previous experiments identified red coloured objects as the most effective at eliciting a behavioural response, demonstrated by increased distance moved within a one-minute interval (unpublished data). All .png objects are available online (see S2).

Predator models and novel object shapes were motion animated in PowerPoint using the *Animation* function. Slide dimensions were 20 x 11 cm, with a standard background image positioned on the lower edge. Predator models were animated with vertical and horizontal movements to simulate a natural swimming motion across the length of the slide, at a speed of ~0.8cm/s (Figure S1C). Novel objects were animated with repetitive horizontal movements at ~0.25cm/s. Template .ppt files were generated to standardise animated motions between the chosen predator models and novel objects respectively. Template .ppt files with additional instructions are accessible online (see S1 - S2). All videos were exported as .wmv files.

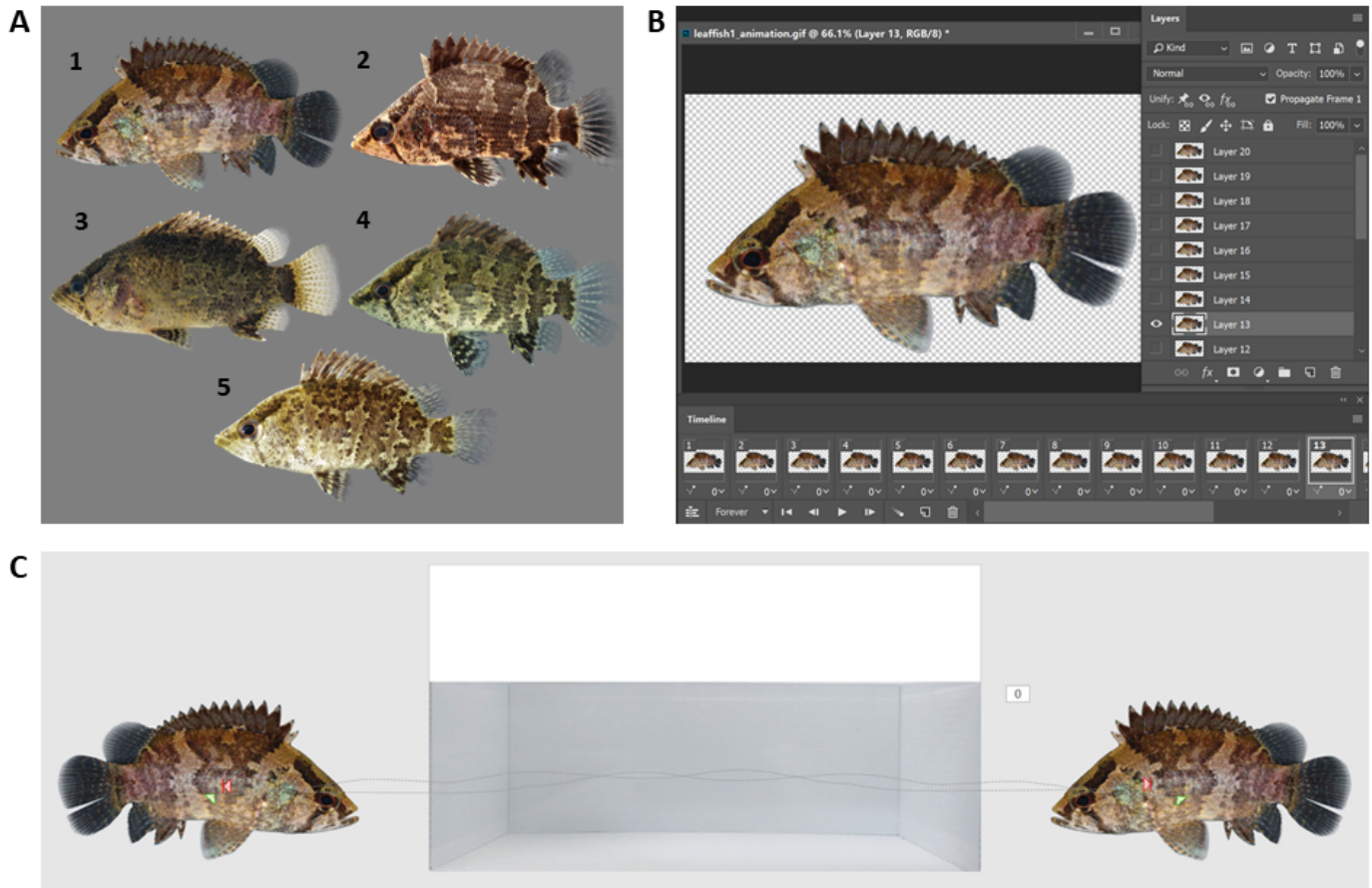


Figure S1: A. Five leaf fish (genus *Nandus*) predator models clipped from photographic images: leaf fish 1 (*Nandus Nebulosus*, [image source](#)), leaf fish 2 (*Nandus oxyrhynchus*, [image source](#)), leaf fish 3 (*Nandus oxyrhynchus*, [image source](#)), leaf fish 4 (*Nandus oxyrhynchus*, [image source](#)) and leaf fish 5 (*Nandus oxyrhynchus*, [image source](#)). **B.** Frame animation clipboard in Photoshop CS6 used to apply waveform animation to each predator model. **C.** Animation template in PowerPoint used to apply motion animation to each predator model. Predator models were animated to 'swim' against a neutral (blank) background.

3.1.4. Generating aggressive and social stimuli

Detailed methods are available online (see S9). Briefly, adult zebrafish were video recorded in a clear acrylic tank lined with white corrugated corflute board. Zebrafish were recorded either individually or in a shoal consisting of five individuals for aggression and social stimulus videos respectively. Collected videos were imported to Photoshop CS6 for video cleaning. Cleaning consisted of re-positioning the captured video within a 20 x 11 cm canvas such that the bottom edge of the tank perfectly aligned with the bottom edge of the canvas. Space above the waterline (7 cm from the bottom of the canvas) was filled in white. Video was exported as .mp4, imported into Windows Live Movie Maker and split into a series of 5-minute stimulus clips. Video clips were exported in .mp4 format.

Exported clips underwent a screening process by three of the authors (MF, SN, RO). Aggression stimulus clips were assessed for aggressive behaviour (evidence of charging, darting or biting behaviour) directed towards or away from the viewer with minimal freezing behaviour. Social stimulus clips were assessed for consistent shoaling behaviour of all

individuals with minimal freezing by one or more individuals. Videos that did not meet pre-defined criteria were excluded. In total, 10 aggression and 8 social preference task clips were generated. All usable aggression and sociability stimulus video clips are available online (see S3 - S4).

3.1.5. Compilation of computer-generated stimuli

Stimulus animations (novel object, predator) and video clips (aggression, social) were compiled into one 38-minute video using Windows Live Movie Maker (WLMM). This video consisted of a 34-minute trial period surrounded by two 2-minute buffer (no stimulus) periods. Videos were designed to play on an infinite loop until all subjects had been trialled for that session. Buffer periods were included to allow zebrafish entry and removal from the experimental tanks by a single operator.

The trial period consisted of 6 phases. Phases 1 and 6, the ‘exploration’ phases, consisted of a 3-minute neutral background image (no stimulus, control). Phases 2 to 5, the ‘stimulus’ phases, consisted of a 3-minute pre-stimulus (no stimulus, control) period, a 3-minute stimulus period, and a 1-minute interval (no stimulus). A series of pilot studies were conducted to determine the optimal period and interval lengths (unpublished data), where the minimum required interval for zebrafish to return to baseline activity measurements was 1-minute. Each stimulus phase comprised one of four behavioural contexts: sociability, neophobia, aggression or boldness (see Figure 1B in the main text).

Periods without a stimulus were created by inserting a neutral ‘blank’ background image (no stimulus) to achieve a seamless transition between each trial phase. This frame was retrieved from video recorded of an empty tank prior to recording the aggression and social preference stimulus videos. For all video phases, captions were inserted in the part of the screen that would be located above the water line to clearly differentiate between the different video phases.

Phases 2 to 5 were pseudorandomised to create 24 possible phase combinations. For each combination, different stimulus animations and video clips were selected to increase variability. Inserted video clips were randomly mirrored using *Visual Effects* to further increase stimulus variability and variate the swimming direction of the predator models. A WLMM template was created to standardise the stimulus phases, and is available online with an accompanying video tutorial (see S5). All videos were published in .mp4 format.

Published videos were imported to Photoshop CS6 for a final rendering step. This step included adjusting video dimensions to match the dimensions of the computer tablets (12 x 7.5 cm), adding labelling features to clearly differentiate each video at the first frame, and adding a timer to monitor video progress during the experimental trials. A Photoshop template with accompanying video tutorial are available online (see S5). All videos were rendered in .mp4 format.

Photoshop rendered videos were compressed using VLC Media Player to improve performance when played on computer tablets. All ready-to-use compressed stimulus videos used in this experiment are available online (see S6).

These methods may be modified for use for alternative video editing programs or computer operating systems. It should be noted that the programs used in the present study were selected due to accessibility and knowledge of use. However, accessibility to WLMM and Photoshop is limited by download availability and cost respectively. Additionally, time taken to render videos in both WLMM and Photoshop was substantial, and may be more so on older operating systems. An alternative free video editing program for Windows that combines the functionality of WLMM and Photoshop CS6 (for video editing) is VSDC Video Editor (available online, [here](#)), which may simplify the video composition process and reduce video rendering time.

3.2. Behavioural phenotyping

3.2.1. Materials used

- 1) Experimental tanks (composed of 6 mm acrylic; 200 x 400 x 120 mm w x l x d, with a white base, white long walls and transparent short walls, Australian Plastic Fabricators Pty Ltd)
- 2) Dark green acrylic panels (composed of 3mm acrylic; 3 x 299 x 125 mm w x l x d); two for every experimental tank
- 3) Heat mat (URS Ultimate Heat Mat, 40 x 60 cm, iPetz); one for every two experimental tanks
- 4) Thermostat (Ringder Digital Aquarium Reptile Dimming Thermostat, DTC-120, 230V, 50HZ, 0-50°C); one for every two heat mats
- 5) Video camcorder (Panasonic HC-V770M camcorder; MP4/720p recording mode); one for every four experimental tanks
- 6) Corrugated white corflute board
- 7) Metal book-ends; one for every experimental tank
- 8) Digital thermometer (Digital Thermometer for Fridge or Freezer IC7209, Instrument Choice); one for every experimental tank
- 9) Computer tablet (Ematic EWT826BK 8-inch Tablet, 1GB Ram, 32GB Storage, Windows 10) connected to OneDrive; two for every experimental tank
- 10) Computer tablet or iOS device with the Panasonic Image App installed; one for every video camcorder
- 11) False-bottom zebrafish breeding containers

3.2.2. Used Windows programs and applications

- 1) Microsoft OneDrive
- 2) Films & TV App (Windows App Store)
- 3) Panasonic Image App (Windows App Store)
- 4) MPEG Streamclip (Squared 5; available online [here](#))

3.2.3. Set up

Experimental tanks and filming equipment was set up according to Figure S2. Briefly, eight experimental tanks were used in this experiment. Each tank was set up with two dark green acrylic panels along the long walls, secured in place with fold-back clips. Tanks were designed with a white floor to heighten subject contrast for data extraction purposes, with transparent acrylic short walls for stimulus video presentation and dark green long walls to minimise anxiety induced by white surroundings³. Each two tanks were positioned on one heat mat whose temperature was controlled by a thermostat. Four tanks were set up in a quadrant formation, with a video camcorder positioned centrally overhead, 104 cm from lens to the base of the tanks. This setup was duplicated to improve high-throughput efficiency, and separated using a sheet of white corflute board. The entire experimental setup was positioned on a custom-made filming rig and surrounded by white corflute board to achieve uniform lighting and minimise visual disturbances during the filming process.

Each experimental tank was filled to 7 cm depth with water, with water temperature maintained at ~28°C throughout the trial. The water temperature of each tank was monitored using a digital thermometer with temperature probe. Water temperature for each tank was recorded at the start of each trial, and the tanks were stirred between trials to minimise temperature gradients.

Stimulus videos were presented to the experimental fish via a series of sixteen computer tablets. Tablets were synchronised using OneDrive such that all tablets had access to all stimulus videos. When in use, computer tablets were inserted into plastic sleeves (to protect from water damage) and positioned flat against the short walls of the experimental tanks, on the outside of the tank (see Figure 1A in the main text). Each tank was set up with two computer tablets; one 'blank' tablet that continuously presented a neutral background image (no stimulus, control), and one 'stimulus' tablet that presented one of the 24 stimulus videos. Blank tablets were always positioned on the outer ends of the experimental tanks (see Figure S2).

3.2.4. Behavioural phenotyping trials

Twenty-four adult zebrafish (male = 12; female = 12, approximately 6 months old) were trialled in this experiment. All subjects had been marked with uniquely coloured elastomer markings for identification, using previously described methods⁴. Behavioural trials occurred between 1 and 4 pm over four days, at weekly intervals, measuring each individual four times over the course of the study.

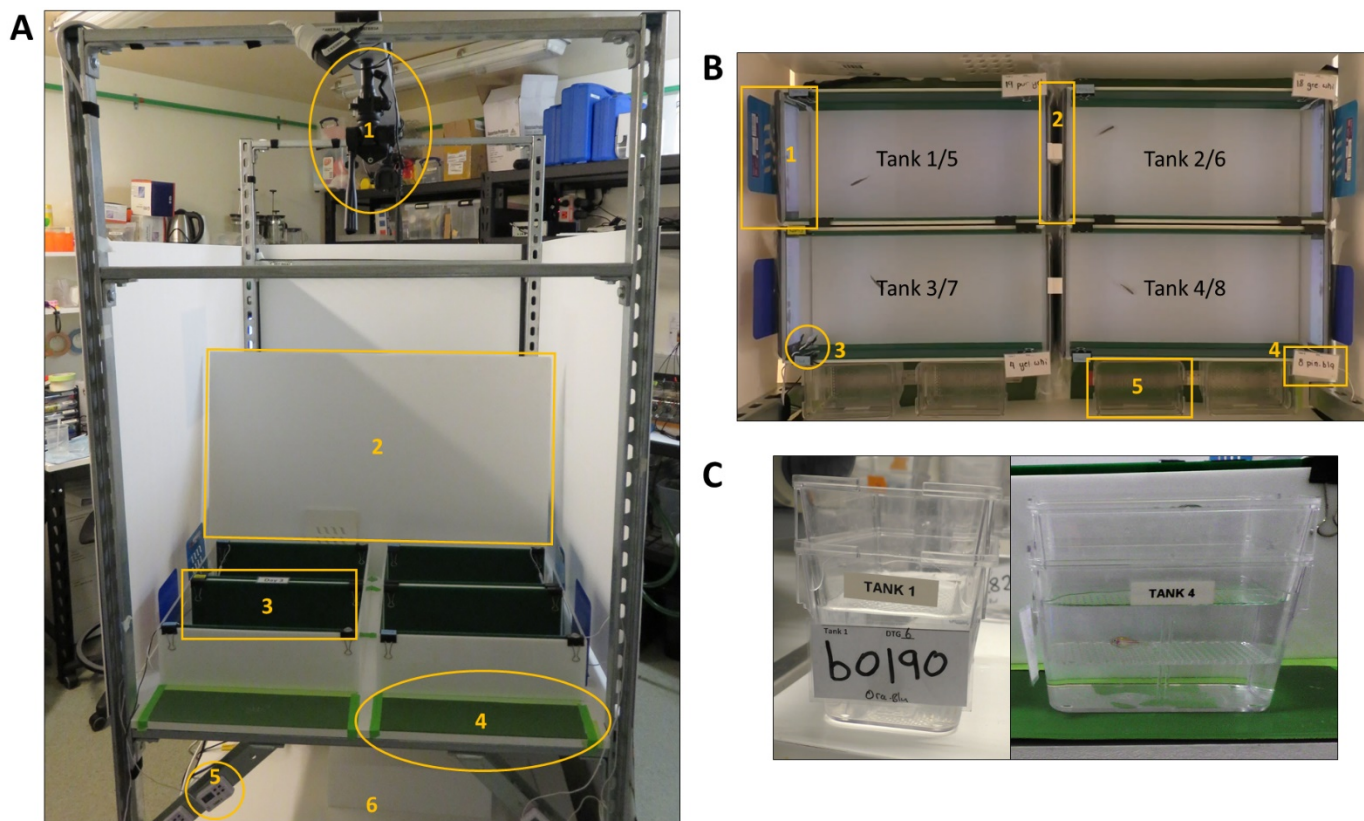
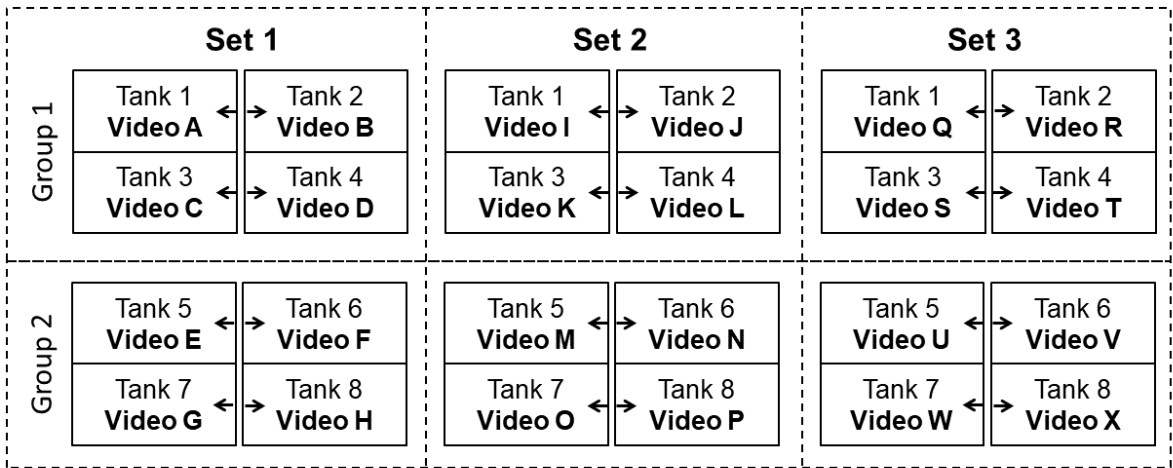


Figure S2: A. Experimental tanks were set up on a custom-built filming rig. Tanks were filmed overhead (1) using a video camcorder. Two tank groups of four were divided by one sheet of white corflute board (2) with the remaining walls of the rig also surrounded by white corflute board. Dark green acrylic panels were clipped to the long walls of the experimental tanks using metal clips (3). Each two heat mats were positioned on a temperature controlled heat mat (4). Digital thermometers were used to monitor to temperature of the experimental tanks during the trials (5). All electrical sources and equipment, including thermostats, were stored on the lower shelf of the filming rig (6). **B.** For each tank group, experimental tanks were set up in a quadrant formation (tank group 1 consisted of tanks 1-4, tank group 2 consisted of tanks 5-8). Blank (no stimulus, control) tablets were positioned against the outer ends of the tanks and were supported with metal book ends (1). Stimulus tablets were positioned against the inner ends of the tanks and were supported with rubber wedges (2). Temperature probes from digital thermometers were secured to the outer corners of the experimental tanks using tape (3) with the temperature probes from the thermostat secured similarly in tank 3 (tank group 1) and tank 7 (tank group 2). Experimental tanks were labelled according to the subject ID for each trial (4). Subjects were held in false bottom breeding containers in front of the experimental tanks prior to trialling (5). **C.** The false bottom breeding containers used to house study subjects prior to trialling, labelled with their unique subject ID and tank number.

The experiment was designed such that each individual was trialled in four unique tanks, and exposed to four unique stimulus videos. Briefly, zebrafish were randomly assigned to one experimental tank on the first trial day. On the following days, zebrafish were moved between experimental tanks in a systematic order. Specifically, $1 > 8 > 3 > 6 > 5 > 4 > 7 > 2$. That is, if an individual was randomly assigned to tank 8 on day one, they would be trialled in tank 3 on day two, and so on. Tank order was designed such that each individual was exposed to both tank groups (tanks 1-4, and tanks 5-8; see Figure S3) on either end of the filming rig, and received stimulus presentation on both the left and right hand sides of the tank in an alternated manner.



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Figure S3: Eight experimental tanks were grouped into two groups: group 1 (tanks 1-4) and group 2 (tanks 5-8). Stimulus videos were allocated to each experimental tank over a series of three sets: set 1 (videos A-H), set 2 (videos I-P) and set 3 (videos Q-X). Arrows indicate the side of stimulus video presentation. Using the methods described herein, an individual randomly assigned to tank 1 set 1 on the first trial day would be assigned to tank 8 set 2 on the second trial day, and so on.

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Additionally, stimulus videos were grouped into three sets (see Figure S3) and assigned to one experimental tank for the duration of the experiment. If an individual was presented with a video from set 1 on the first trial day, they would be presented with a video from set 2 on the second trial day, and so on. In this manner, an individual could be exposed to up to 24 unique stimulus videos without repeated exposure. Thus, this experiment was designed to be expandable to larger sample sizes and repetitions.

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For each trial, stimulus videos were played synchronously using the Films & TV App (Windows 10) for each tank group. Video recording was controlled remotely using Panasonic Image App on a computer tablet. Recording commenced at 00:00 of the stimulus videos, and ended at 36:10.

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Zebrafish were held in individual holding containers prior to their trial (Figure S2). For each trial, fish were placed individually into an experimental tank during an initial 2 min buffer period. Zebrafish were exposed to a 34-minute trial period (see Figure 1 in the main text), and then removed from their tanks and returned to their holding containers. Computer tablets were programmed to cycle through a predetermined stimulus video on a loop, allowing subjects to be trialled in succession in an efficient and high-throughput manner. Therefore, for each trial day, the selected video set did not change. Using these methods, up to 64 subjects could be easily trialled in a 6-hour period. Detailed methods, including an instructional video tutorial outlining this process are available online (see S10 - S11).

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Preliminary data was collected in an excel spreadsheet and transferred to EthoVision during data acquisition (available online, S7).

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For this experiment, collected video files were renamed according to the trial day, filming session, and camera number (i.e. d1_s1_c1). Due to restrictions with our chosen video

camcorder, videos were broken into two clips (one 30-minute clip, and one 6-minute clip). Videos were concatenated using MPEG Streamclip with audio removed for analysis in EthoVision.

3.2.5. EthoVision

Video footage was analysed using Noldus EthoVision 11.5 XT. An EthoVision template consisting of sample videos, arena settings, trial control settings, detection settings, data profiles and analysis profiles can be found online (see S8) with detailed methods (S12). Methods may be adapted to utilise alternative, freely available animal tracking software e.g. ⁵

4. Statistical analyses and extended results

To ascertain whether traits show consistent individual differences, we quantified individual repeatability (i.e., intraclass correlation coefficient - ICC) for each behavioral trait of interest using the *rprr* function in the package *rprr* ⁶ in the R statistical environment (version 3.4.1) ⁷. We choose four independent variables that we deemed, *a priori*, to be biologically relevant measures of behaviour in zebrafish, and that were least correlated with each other [speed of movement (cm/s); distance to stimulus (cm); duration along edge (s) and distance moving (s)]. Distance to stimulus (in Social and Aggression contexts) and duration along edge (in Novel and Predatory contexts) were most clearly affected when the stimulus was conveyed on the tablets (Figure S4). Additional variables that quantify the change in behaviour between pre-stimulus and stimulus periods could also be computed with our design and repeatability of these traits can also be quantified. Nonetheless, we were interested in understanding, more generally, patterns of repeatability in these traits and so we only utilized the raw behaviours. We calculated unadjusted repeatability (i.e., not accounting for fixed effects) for each variable using a mixed model framework ⁶. Confidence intervals were generated by conducting 1000 parametric bootstraps. Sex can be an important source of individual differences, and yet, while accounting for sex did lead to decreased repeatability, many traits remained significantly repeatable across contexts (See “Adjusted repeatability” - Figure S5).

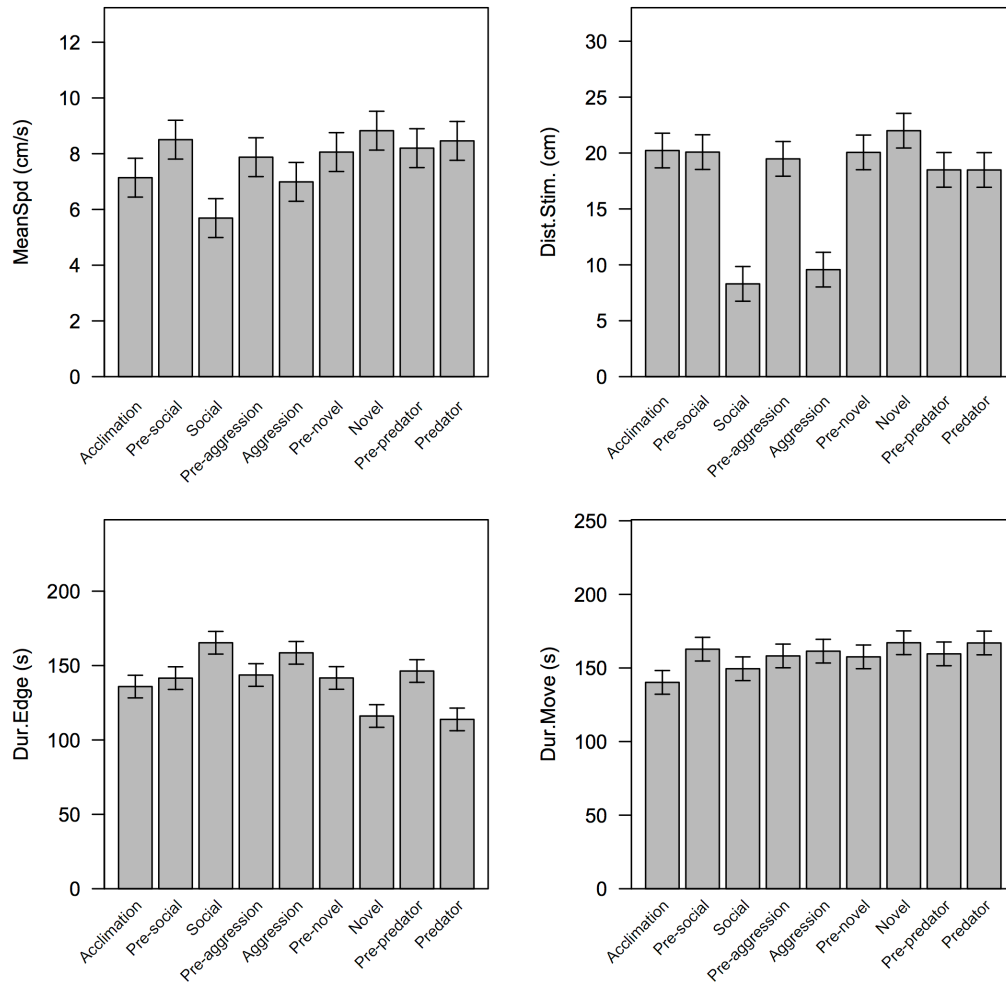


Figure S4: Swimming speed (cm/s), distance to stimulus (cm), time on edge (s) and duration moving (s) for the various experimental periods across individuals. Mean (+/- standard error) are provided.

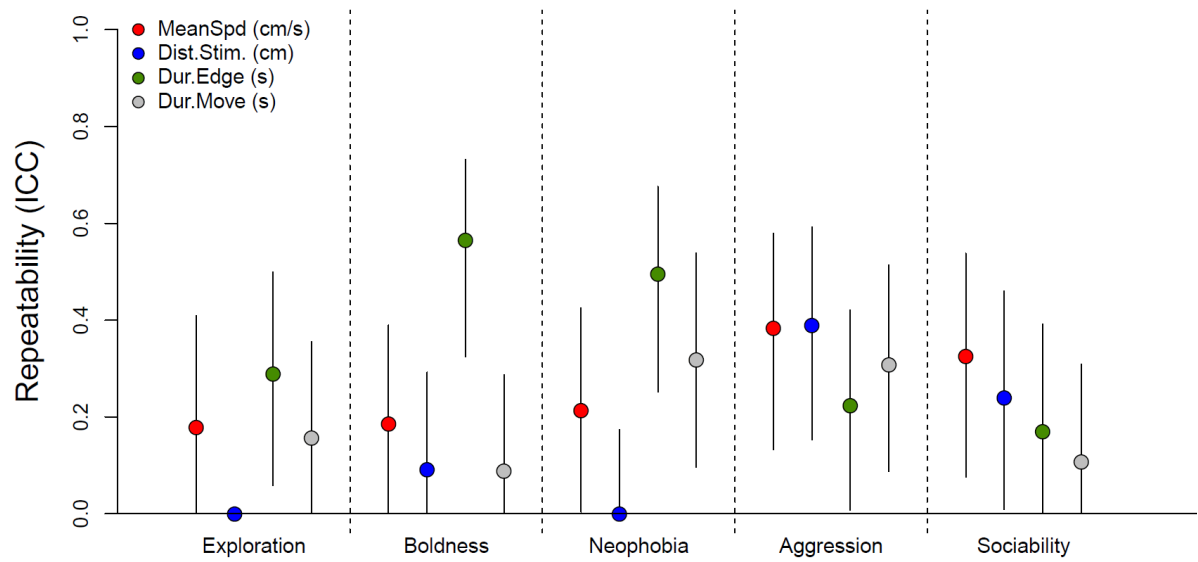


Figure S5: Adjusted repeatability estimates and their 95% confidence intervals after accounting for sex effects. Significant repeatability is signified by a lack of overlap between 95% CI's and zero.

6. References

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